

ECA0303-SIGNALS AND SYSTEM

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Experiment 1.1

Generation of Continuous Time Signals

Program:

% Generation of Continuous Time Signals

clc;

clear;

close all;

% Time axis

t = -5:0.01:5;

% Continuous-time sine signal

x = sin(2*pi*t);

% Plotting

figure;

plot(t, x, 'LineWidth', 2);

grid on;

title('Continuous Time Sine Signal');

xlabel('Time (seconds)');

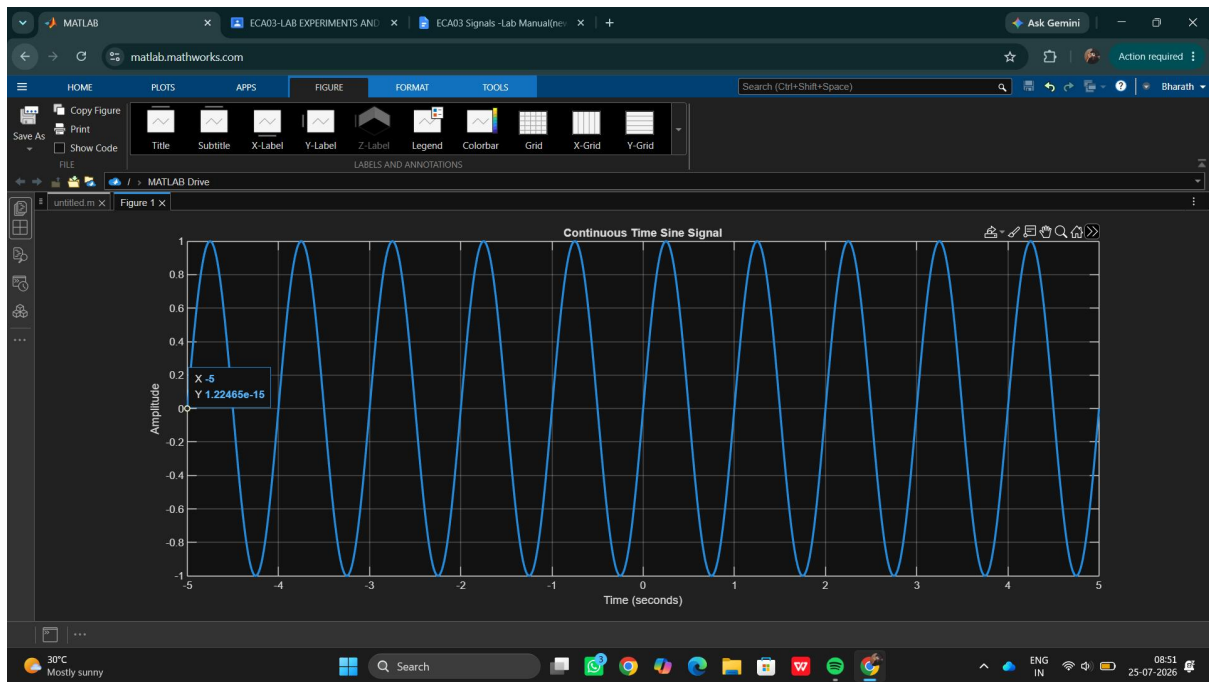
ylabel('Amplitude');

Expected Output:

A continuous-time sine wave is displayed with:

- Time range: -5 s to 5 s
- Amplitude range: -1 to 1
- Periodic sinusoidal waveform

Output Waveform:



Result:

This test case validates that the signal generator can produce a sinusoidal waveform with the specified frequency, amplitude, and phase shift.